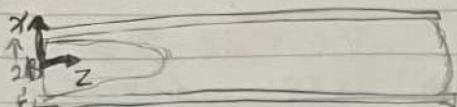


Homework 4



viscosity μ

constant density ρ

distance $2B$

@ $t=0$ $\frac{\Delta P}{L} = \frac{p(z=L) - p(z=0)}{L}$

- Using the relevant equation of motion show that the velocity profile is governed by the following PDE:

$$\rho \frac{dv_z}{dt} = \mu \frac{d^2 v_z}{dx^2} - \frac{\Delta p}{L}$$

$v_z(t, x)$

$$\rho \frac{dv_z}{dt} = - \frac{dp}{dz} + \mu \frac{d^2 v_z}{dx^2}$$

$$- \frac{dp}{dz} = \frac{\Delta p}{L}$$

$$\rho \frac{dv_z}{dt} = \mu \frac{d^2 v_z}{dx^2} - \frac{\Delta p}{L}$$

- No slip boundary conditions, demonstrate steady-state velocity profile.

$v_{z\infty} = v_{max} \left[1 - \left(\frac{x}{B} \right)^2 \right]$, where $v_{max} = - \frac{B^2 \Delta p}{4\mu L}$

B.C

$v_{z\infty}(x=B) = 0$

$v_{z\infty}(x=B) = 0$

$v_{z\infty}(x=0) = v_{zmax}$

$\rho \frac{dv_z}{dt} = \mu \frac{d^2 v_z}{dx^2} - \frac{\Delta p}{L}$

$\int \int \frac{\Delta p}{L} = \mu \frac{d^2 v_z}{dx^2}$

$\mu \frac{dv_z}{dx} = \frac{\Delta p}{L} x + C_1 \Rightarrow \mu v_{z\infty} = \frac{\Delta p}{2L} x^2 + C_1 x + C_2$

$0 = \frac{\Delta p}{2L} B^2 + C_1 B + C_2$

$0 = \frac{\Delta p}{2L} B^2 + C_1 B + C_2$

$v_{z\infty} = \frac{\Delta p}{4\mu L} x^2 + \frac{\Delta p B^2}{4\mu L}$

$0 = \frac{\Delta p}{L} B^2 + 2C_2$

$C_2 = - \frac{\Delta p B^2}{2\mu L}$

$C_1 = 0$

$v_{z\infty} = - \frac{B^2 \Delta p}{4\mu L} \left[1 - \left(\frac{x}{B} \right)^2 \right]$

3- Demonstrate part 1 can be expressed in a dimensionless form: $\frac{d\phi}{d\tau} = \frac{d^2\phi}{d\eta^2}$ where $\phi = \frac{v_z}{v_{max}}$, $\tau = \frac{\mu t}{\rho B^2}$, $\eta = \frac{x}{B}$

$$\rho \frac{dv_z}{dt} = \mu \frac{d^2 v_z}{dx^2} - \frac{\Delta P}{L}$$

$$\frac{\rho v_z / v_{max}}{dt \cdot \frac{\mu}{\rho B^2}} = \mu \frac{d^2 v_z / v_{max}}{dx^2 \cdot \frac{\mu}{\rho B^2}} - \frac{\Delta P / v_{max}}{L \cdot \frac{\mu}{\rho B^2}}$$

$$\frac{\rho}{\rho} \frac{d\phi}{d\tau} = \mu \frac{\rho \phi}{\mu \eta^2} \cdot \frac{\rho}{\rho} + \Delta P \cdot \frac{2\mu K}{B^2 \Delta P} \cdot \frac{\rho B^2}{K \mu}$$

* divide by S

$$\frac{d\phi}{d\tau} = \frac{d^2\phi}{d\eta^2} + 2$$

$v_{max} = \frac{\rho B^2 \Delta P}{2\mu L}$

4- Steady State Velocity profile is form $\phi = 1 - \eta^2$
 $v_{z0} = v_{max} [1 - (\frac{x}{B})^2]$

$$\phi = \frac{v_z}{v_{max}}$$

$$\phi = 1 - \frac{x^2}{B^2}$$

$$\phi = 1 - \eta^2$$

$$\frac{x^2}{B^2} = \eta^2$$

* 5- $\phi = \phi(\eta, \tau)$ write down I.C and two BC

IC $\phi(\eta, \tau=0) = 0$

BC $\phi(\eta=-1, \tau) = 0$

BC $\phi(\eta=+1, \tau) = 0$

$$v_z(x=B, t) = 0$$

$$\frac{dv_z}{dx} \left(x = \frac{B}{2}, \tau \right) = 0$$

$$\phi(\eta=1, \tau) = 0$$

$$6- \phi(\eta, \tau) = \sum_{n=1}^{\infty} C_n(\tau) \cos\left(\frac{(2n-1)\pi\eta}{2}\right)$$

Sub into $\Rightarrow \frac{d\phi}{d\tau} = \frac{d^2\phi}{d\eta^2} + 2$
 derive ODE eqn. for $C_n(\tau)$ evaluate integration constants using the proper orthogonality eqn. for the cosine function.

$$\frac{d\phi}{d\tau} = \frac{d^2\phi}{d\eta^2} + 2$$

B.C.
 $\eta=0 \quad \frac{d\phi}{d\eta} = 0 \quad \phi=1$
 $\eta=1 \quad \phi=0$

$$\frac{d\phi}{d\tau} = \sum_{n=1}^{\infty} \frac{d}{d\tau} [C_n(\tau) \cos(\frac{(2n-1)\pi\eta}{2})]$$

$$\frac{d\phi}{d\tau} = \sum_{n=1}^{\infty} -C_n(\tau) \sin(\frac{(2n-1)\pi\eta}{2}) \cdot (\frac{(2n-1)\pi}{2})$$

$$\frac{d^2\phi}{d\eta^2} = \sum_{n=1}^{\infty} -C_n(\tau) \cos(\frac{(2n-1)\pi\eta}{2}) \cdot (\frac{(2n-1)\pi}{2})^2$$

$$\sum_{n=1}^{\infty} \frac{d[C_n(\tau) \cos(\frac{(2n-1)\pi\eta}{2})]}{d\tau} = - \sum_{n=1}^{\infty} C_n(\tau) \cos(\frac{(2n-1)\pi\eta}{2}) \cdot (\frac{(2n-1)\pi}{2})^2 + 2$$

$$\sum_{n=1}^{\infty} \frac{d}{d\tau} [C_n(\tau) \cos(\frac{(2n-1)\pi\eta}{2}) \cos(\frac{(2n-1)\pi\eta}{2})] = - \sum_{n=1}^{\infty} C_n(\tau) \int_{-1}^1 \cos(\frac{(2n-1)\pi\eta}{2}) \cos(\frac{(2n-1)\pi\eta}{2}) \cdot (\frac{(2n-1)\pi}{2})^2 d\eta$$

$$\sum_{n=1}^{\infty} \frac{dC_n(\tau)}{d\tau} = - \sum_{n=1}^{\infty} C_n(\tau) (\frac{(2n-1)\pi}{2})^2 + \int_{-1}^1 2 \cos(\frac{(2n-1)\pi\eta}{2})$$

$$\sum_{n=1}^{\infty} \frac{dC_n(\tau)}{d\tau} = \sum_{n=1}^{\infty} C_n(\tau) (\frac{(2n-1)\pi}{2})^2 - \left(\frac{2 \sin(\frac{(2n-1)\pi}{2})}{\frac{(2n-1)\pi}{2}} \right) \frac{2 \sin(\frac{(2n-1)\pi}{2})}{\frac{(2n-1)\pi}{2}}$$

$$\frac{dC_n(\tau)}{d\tau} = -C_n(\tau) (\frac{(2n-1)\pi}{2})^2 - \dots$$

$$N=1 \quad \frac{2 \sin(\frac{1}{2}\pi)}{\frac{1}{2}\pi} - \frac{2 \sin(\frac{1}{2}\pi)}{\frac{1}{2}\pi} + 2$$

$$\frac{dC_1(\tau)}{d\tau} = -C_1(\tau) (\frac{(2N-1)\pi}{2})^2 = \frac{(1-\pi)^2}{(1-\pi)\pi}$$

$$\frac{dC_N}{dx} = -C_N(x) \cdot \left(\frac{2n-1}{2}\pi\right)^2 - (-1)^n 8 / (n-1)\pi$$

$$\frac{dC_N}{dx} + \left(\frac{2n-1}{2}\pi\right)^2 C_N(x) = -\frac{(-1)^n 8}{(n-1)\pi}$$

$$u(x) = e^{\int \frac{2n-1}{2}\pi dx} = e^{\frac{(2n-1)\pi}{2}x}$$

$$e^{\frac{(2n-1)\pi}{2}x} \frac{dC_N}{dx} + \left(\frac{2n-1}{2}\pi\right)^2 e^{\frac{(2n-1)\pi}{2}x} C_N = -\frac{(-1)^n 8}{(n-1)\pi} e^{\frac{(2n-1)\pi}{2}x}$$

$$\int \frac{d}{dx} e^{\frac{(2n-1)\pi}{2}x} C_N dx = \int -\frac{(-1)^n 8}{(n-1)\pi} e^{\frac{(2n-1)\pi}{2}x} dx$$

$$e^{\frac{(2n-1)\pi}{2}x} C_N = -\frac{(-1)^n 8}{(n-1)\pi} \cdot \frac{1}{\frac{(2n-1)\pi}{2}} e^{\frac{(2n-1)\pi}{2}x} + C$$

$$C_N = -\frac{(-1)^n 8}{(n-1)\pi} \cdot \frac{2}{(2n-1)\pi} + B / e^{\frac{(2n-1)\pi}{2}x}$$

$$C_N = B \exp\left(-\frac{(2n-1)\pi}{2}x\right) - \frac{(-1)^n 32}{[(2n-1)\pi]^3}$$

BC

$$\phi(\eta, x=0) = 0$$

$$= \sum_{n=1}^{\infty} \left(B_n - \frac{(-1)^n 32}{[(2n-1)\pi]^3} \right) \cos\left(\frac{2n-1}{2}\pi\eta\right)$$

$$\rightarrow B_n = \frac{(-1)^n 32}{[(2n-1)\pi]^3}$$

